



Student Name: _____

Teacher: _____

2023

Hurlstone Agricultural High
School
Year 12 Assessment Task 2

Mathematics Extension 2

Examiners

- Mr D Potaczala
- Mr G Huxley
- Mr G Rawson

General Instructions

- Reading time – 3 minutes
- Working time – 40 minutes
- Write using black or blue pen
- NESAs-approved calculators may be used
- A Reference sheet is provided for your use
- An A4 handwritten summary sheet may be used
- In Questions 7 to 9, show relevant mathematical reasoning and/or calculations

Total marks: 30

Section I – 6 marks (page 2–3)

- Attempt Questions 1–6
- Allow about 8 minutes for this section

Section II – 24 marks (pages 4–6)

- Attempt Questions 7–9
- Allow about 32 minutes for this section
- For each question use a separate answer booklet

Section I

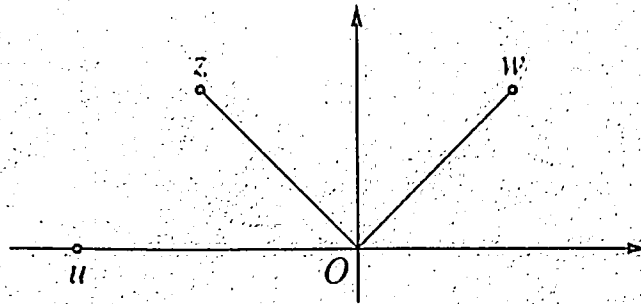
6 marks

Attempt Questions 1 to 6

Allow about 8 minutes for this section

Use the multiple-choice answer sheet for Questions 1 to 6

1. The argand diagram shows the complex numbers w , z , and u , where w lies in the first quadrant, z lies in the second quadrant and u lies on the negative real axis.



Which statement could be true?

- A. $u = zw$ and $u = z + w$ B. $u = zw$ and $u = z - w$
- C. $z = uw$ and $u = z + w$ D. $z = uw$ and $u = z - w$
2. It is known that a particular complex number z is NOT a real number.
- Which of the following could be true for this number z ?
- A. $\bar{z} = iz$ B. $\bar{z} = |z^2|$
- C. $\operatorname{Re}(iz) = \operatorname{Im}(z)$ D. $\operatorname{Arg}(z^3) = \operatorname{Arg}(z)$
3. Which complex number lies in the region $2 < |z - 1| < 3$?

A. $1 + \sqrt{3}i$

B. $1 + 3i$

C. $2 + i$

D. $3 - i$

4. Given that $z = 1 - i$, which expression is equal to z^3 ?

A. $\sqrt{2} \left(\cos \left(\frac{-3\pi}{4} \right) + i \sin \left(\frac{-3\pi}{4} \right) \right)$

B. $2\sqrt{2} \left(\cos \left(\frac{-3\pi}{4} \right) + i \sin \left(\frac{-3\pi}{4} \right) \right)$

C. $\sqrt{2} \left(\cos \left(\frac{3\pi}{4} \right) + i \sin \left(\frac{3\pi}{4} \right) \right)$

D. $2\sqrt{2} \left(\cos \left(\frac{3\pi}{4} \right) + i \sin \left(\frac{3\pi}{4} \right) \right)$

5. Which statement is true?

A. $\forall x \in \mathbb{R}, \exists y \in \mathbb{R}$ such that $xy = 6$.

B. $\forall x \in \mathbb{R}, \exists y \in \mathbb{R}$ such that $x + y = 6$.

C. $\exists x \in \mathbb{R}$ such that $\forall y \in \mathbb{R}, x + y = 6$.

D. $\exists x \in \mathbb{R}$ such that $\forall y \in \mathbb{R}, xy = 6$.

6. Consider the statement P .

P : For all integers $n \geq 1$, if n is a prime number then $\frac{n(n+1)}{2}$ is a prime number.

Which of the following is true about this statement and its converse?

A. The statement P and its converse are both true.

B. The statement P and its converse are both false.

C. The statement P is true and its converse is false.

D. The statement P is false and its converse is true.

~ END OF SECTION I ~

Section II

18 marks

Attempt Questions 7 to 9

Allow about 32 minutes for this section

Answer each question in a new answer booklet.

All necessary working should be shown in every question.

Question 7 (8 marks)

Marks

- (a) Let $z = 5 - i$.
- (i) Find $z + 2\bar{z}$. 1
- (ii) Find $\frac{i}{z}$ in the form $x + iy$. 1
- (b) Prove that $2i$ is a root of $x^3 - 3x^2 + 4x - 12 = 0$. Hence, find all other roots. 3
- (c) Given the complex number $z = e^{i\theta}$, show that $w = \frac{z^2 - 1}{z^2 + 1}$ is purely imaginary. 3

~ End of question 7 ~

Question 8 (8 marks)

Marks

- (a) (i) Sketch the region on the Argand diagram where the following inequalities hold simultaneously: 3

$$|z - 1| < \sqrt{2} \text{ and } 0 \leq \arg(z + i) \leq \frac{\pi}{4}$$

- (ii) Let w be the complex number of minimum modulus satisfying the inequalities in part (i). Express w in the form $x + iy$, where x and y are real number. 1

- (b) Let $z = \cos \theta + i \sin \theta$.

- (i) Show that $z^n - \frac{1}{z^n} = 2i \sin n\theta$. 2

- (ii) Using the expansion of $\left(z - \frac{1}{z}\right)^5$, show that: 2

$$\sin^5 \theta = \frac{\sin 5\theta - 5 \sin 3\theta + 10 \sin \theta}{16}$$

~ End of question 8 ~

Question 9 (8 marks)

Marks

- (a) Consider the following statement, which is of the form $P \rightarrow Q$.

"If the pet is a stallion, then it must be a horse."

- | | | |
|-------|--|---|
| (i) | Write the converse of the statement. | 1 |
| (ii) | Write the contrapositive of the statement. | 1 |
| (iii) | Write the negation of the statement | 1 |

- (b) Write a proof by contradiction to show that: 2

There are no integers a and b such that $21a + 3b = 1$.

- (c) (i) Write an inequality that describes the statement: 1

The area of a rectangle cannot be greater than the square of the average length of its two sides.

- (ii) Prove the statement in part (i).

~ END OF SECTION II ~

~ END OF PAPER ~

1)

Since z and w have positive imaginary components $u = z - w$ can be true as the imaginary parts can cancel to give only a real.

Also, can be true $u = zw$ as the addition of an acute and obtuse argument can result in a straight angle.

Option B

2)

Since z is not real, we have either $z = x + yi$ or $z = yi$

B is not an option as the modulus is a real number

C is not an option as $\operatorname{Re}(z) = 0$ is a possibility.

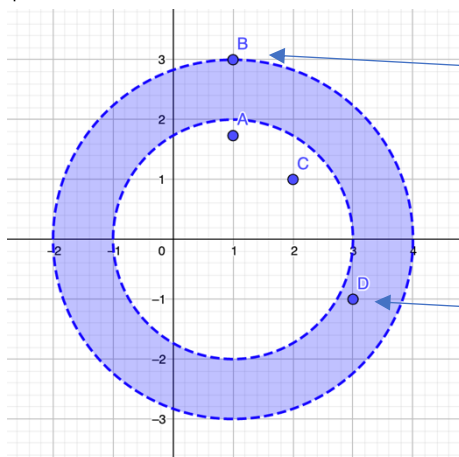
D $\arg(r^3 e^{3i\theta}) = \arg(re^{i\theta}) \rightarrow 3i\theta = i\theta$ which is not true

Hence option A must be true ($\bar{z} = iz$ is true if $z = y + yi$).

Option A

3)

$$2 < |z - 1| < 3$$



It *can't* be B, because B is on the boundary, which is not included.

That's your answer, there.

D

Option D

4)

$$\begin{aligned}
 z^3 &= (1 - i)^3 \\
 &= \left(\sqrt{2} e^{-i\frac{\rho}{4}} \right)^3 \\
 &= 2\sqrt{2} e^{-i\frac{3\rho}{4}} \\
 &= 2\sqrt{2} \left(\cos\left(\frac{-3\rho}{4}\right) + i \sin\left(\frac{-3\rho}{4}\right) \right)
 \end{aligned}$$

Option B

5)

Interpretations in words:

A. For all real x , there is a y -value such that $xy=6$. This is false, with $x=0$ a counter-example.

B. For all real x , there is a y -value such that $x + y = 6$. This is true.

C. There exists a real x -value that has all y -values satisfying $x + y=6$. This is false because every x -value has a single unique y -value that makes the equation true.

D. There exists a real x -value that has all y -values satisfying $xy=6$. This is false, because every x -value in the domain has a single unique y -value that makes the equation true.

Option B

6)

The 4 response choices consider the combinations of the statement and its converse. Consider each statement:

Statement P is false. When testing prime numbers, test the number 2 (the only even prime) and an odd prime. $n=3$ is a counter-example to the statement.

When testing statement P you may have noticed that $n(n+1)$ is always an even product of two factors. The only time the numerator is fully factorised by the denominator is when $n=2$.

Hence, the only time the fraction is prime is when $n=2$.

The converse: "If the fraction is prime, then n is prime" is true.

Option D

2023 Yr12 HSC Assessment Task 2	
Question 7	Solutions and Marking Guidelines
Outcomes Addressed in this Question	
MEX12-4 uses the relationship between algebraic and geometric representations of complex numbers and complex number techniques to prove results, model and solve problems	
Solutions	Marking Guidelines
a i) $z + 2\bar{z} = 5 - i + 2(5 + i)$ $= 15 + i$ ii) $\frac{i}{z} = \frac{i}{5 - i} = \frac{(5 + i)i}{25 + 1}$ $= \frac{-1 + 5i}{26} = -\frac{1}{26} + \frac{5}{26}i$ b) $f(2i) = (2i)^3 - 3(2i)^2 + 4(2i) - 12$ $= -8i + 12 + 8i - 12$ $= 0$ By factor theorem $2i$ is a root. Since the coefficients of the polynomial are real, the complex roots come in conjugate pairs. Hence $-2i$ is another root. Summing the roots we have: $2i - 2i + \alpha = -\frac{-3}{1}$ $\alpha = 3$	1 mark Correct solution 1 mark Correct solution 3 marks Correct solution with clear reasoning 2 marks Substantial progress towards solution 1 mark Some progress towards solution.

c)

$$w = \frac{z^2 - 1}{z^2 + 1}$$

$$= \frac{z \left(z - \frac{1}{z} \right)}{z \left(z + \frac{1}{z} \right)}$$

$$= \frac{e^{i\theta} - e^{-i\theta}}{e^{i\theta} + e^{-i\theta}}$$

$$= \frac{\cos\theta + i\sin\theta - \cos\theta + i\sin\theta}{\cos\theta + i\sin\theta + \cos\theta - i\sin\theta}$$

$$= \frac{2i\sin\theta}{2\cos\theta}$$

Which is imaginary.

3 marks
Correct solution with
clear reasoning

2 marks Substantial
progress towards
solution

1 mark
Some progress
towards solution.

Year 12	Mathematics Extension 2	Ass Task 2 2023 HSC
Question No. 8	Solutions and Marking Guidelines	
Outcomes Addressed in this Question		
MEX12-4	uses the relationship between algebraic and geometric representations of complex numbers and complex number techniques to prove results, model and solve problems	
Part / Outcome	Solutions	Marking Guidelines
(a)(i)		<u>3 marks</u> : correct solution Note: the open circle at (0, -1) is necessary for full marks. <u>2 marks</u> : substantially correct solution <u>1 mark</u> : partially correct solution
(a)(ii)	For w to lie in the region in part (i) and be of minimum modulus, w must be in the position as shown below... ie the perpendicular to $y = x - 1$, through O. By inspection, $w = \frac{1}{2} - \frac{1}{2}i$	<u>1 mark</u> : correct solution
(b)(i)	$z^n - \frac{1}{z^n} = z^n - z^{-n}$ $= (\cos q + i \sin q)^n - (\cos q + i \sin q)^{-n}$ $= \cos nq + i \sin nq - (\cos(-nq) + i \sin(-nq)) \quad (\text{de Moivre})$ $= \cos nq + i \sin nq - (\cos nq - i \sin nq)$ $= 2i \sin nq$	<u>2 marks</u> : correct solution <u>1 mark</u> : substantially correct solution

(b)(ii)	$\left(z - \frac{1}{z}\right)^5 = z^5 - 5z^4 \frac{1}{z} + 10z^3 \frac{1}{z^2} - 10z^2 \frac{1}{z^3} + 5z \frac{1}{z^4} - \frac{1}{z^5}$ $= z^5 - 5z^3 + 10z - 10\left(\frac{1}{z}\right) + 5\left(\frac{1}{z^3}\right) - \frac{1}{z^5}$ $= \left(z^5 - \frac{1}{z^5}\right) - 5\left(z^3 - \frac{1}{z^3}\right) + 10\left(z - \frac{1}{z}\right)$ so, $(2i \sin q)^5 = 2i \sin 5q - 5(2i \sin 3q) + 10(2i \sin q)$ $32i \sin^5 q = 2i \sin 5q - 10i \sin 3q + 20i \sin q$ $= \frac{2i \sin 5q - 10i \sin 3q + 20i \sin q}{32i}$ $= \frac{\sin 5q - 5 \sin 3q + 10 \sin q}{16}$	<u>2 marks</u> : correct solution <u>1 mark</u> : substantially correct solution
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Year 12	Mathematics Extension 2	Ass Task 2 2023 HSC
Solutions and Marking Guidelines Question 9		
Outcomes Addressed in this Question		
E2: Chooses appropriate strategies to construct arguments and proofs in both concrete and abstract settings.		
Part	Solutions	Marking Guidelines
Q9 (a)	(a) The question has a $P \Rightarrow Q$ statement. (i) Converse has the form: $Q \Rightarrow P$ Hence: “If the pet is a horse, it must be a stallion.” (ii) Contrapositive has the form: $\neg Q \Rightarrow \neg P$. Hence: “If the pet is not a horse, it is not a stallion.” (iii) Negation has the form $\neg P \Rightarrow \neg Q$. Hence: “If not a stallion, then the pet is not a horse.” Negations can also replace universal statements with “there exists”. Hence the alternative: “There exists a stallion that is not a horse.”	(a)(i)1 mark: correct statement. (ii)1 mark: correct statement. (iii)1 mark: correct statement.
	(b) (b) Assume that a and b are integers, and $21a + 3b = 1$ Then: $3(7a + 3b) = 1$ $7a + b = \frac{1}{3}$ But LHS is an integer for integers a and b. RHS is not an integer, and so we have a contradiction. Hence, the assumption is untrue; both a and b can't be integers.	(b)2 marks: Proper set up and proof by contradiction statement included. 1 mark: One component of proof incomplete.

(c)	(c) (i) Let the length and width of a rectangle be x, y. Inequality will be: $xy \leq \left(\frac{x+y}{2}\right)^2$ (ii) <u>Method 1: Consider the difference:</u> $ \begin{aligned} xy - \left(\frac{x+y}{2}\right)^2 &= xy - \frac{x^2 + 2xy + y^2}{4} \\ &= \frac{4xy - x^2 - 2xy - y^2}{4} \\ &= \frac{-x^2 + 2xy - y^2}{4} \\ &= -\frac{(x-y)^2}{4} \\ &\leq 0 \text{ due to the square being non-negative.} \end{aligned} $ $\therefore xy \leq \left(\frac{x+y}{2}\right)^2$ <u>Method 2: Using the AM/GM inequality.</u> AM/GM: $\frac{a+b}{2} \geq \sqrt{ab}$ square both sides $\left(\frac{a+b}{2}\right)^2 \geq ab$ Which is equivalent to the statement in (i). <i>Note: Method 2 begins with a factual statement as its “axiom”, then progresses towards the statement that needs to be proved.</i>	(c)(i) 1 mark: Correct answer (ii) 2 marks: Proper set up and proof. 1 mark: One component of proof incomplete.
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